

Fault Tolerance

08 June 2026 11:44 AM

Fault tolerance is the ability of a system to **continue working even when some components fail**.

For example:

- One server crashes → Application still works.
- One database goes down → Another database serves requests.
- One data center fails → Traffic is redirected to another.

A fault-tolerant system aims for **high availability** and **minimal downtime**.

Why is Fault Tolerance Important?

Imagine building:

- WhatsApp
- Instagram
- Netflix
- Banking systems

If a single server failure causes the entire system to stop, millions of users are affected.

So HLD systems are designed to remove **Single Points of Failure (SPOF)**.

Techniques for Fault Tolerance

1. Redundancy

Keep multiple copies of components.

Example:

App Server 1

App Server 2

App Server 3

If one dies, others continue.

Types

Active-Active

All servers handle requests.

LB
/ | \
App1 App2 App3

Advantages:

- Better performance
- High availability

Active-Passive

One server works, another waits.

Primary ----> Running

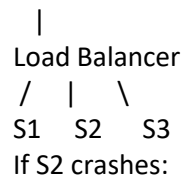
Backup -----> Standby

If primary fails, backup takes over.

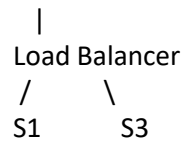
2. Load Balancer

Distributes traffic.

Users



Users



No downtime.

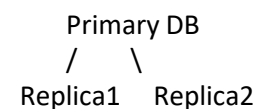
Popular:

- Nginx
- HAProxy
- AWS ELB

3. Replication

Store multiple copies of data.

Database Replication



Writes:

- Primary

Reads:

- Replicas

If primary fails:

- Replica becomes primary.

4. Failover

Automatically switch to backup.

Example:

Primary Server

X (Fails)

Backup Server starts

Failover Types:

- Automatic
- Manual

5. Data Backup

Keep copies of important data.

Example:

- Daily backup
- Hourly backup
- Incremental backup

Without backup:

- Data loss is permanent.

6. Data Partitioning (Sharding)

Instead of one huge database:

Shard1 : User 1-1M

Shard2 : User 1M-2M

Shard3 : User 2M-3M

If one shard fails:

- Entire system may still partially work.

7. Distributed Systems

Instead of one machine:

Node1

Node2

Node3

Node4

Data stored across multiple nodes.

Example:

- Cassandra
- DynamoDB
- HDFS

8. Heartbeat Mechanism

Servers continuously send health signals.

Server ---> "I'm Alive"

Server ---> "I'm Alive"

(No signal)

Monitor ---> Server Failed

Used in:

- Kubernetes
- Distributed databases